Last name _____

First name _____

LARSON—MATH 310–CLASSROOM WORKSHEET 18
QR factorization and Inverses.

Review

1. Find A^2, A^3.

QR Factorization (Sec. 10.4)
Let

$$A = \begin{bmatrix} 0 & 1 & 2 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

1. Let the columns of A be $\vec{a}_1, \ldots, \vec{a}_6$. Use the **Gram-Schmidt algorithm** to find orthonormal $\vec{q}_1, \ldots, \vec{q}_r$ (where r is the number of produced $\vec{q}$ vectors). Write each $\vec{a}_i$ in terms of $\vec{q}_i$'s. Note that is a column $\vec{a}_i$ is not linearly independent from the previously produced $\vec{q}$'s, it can be written in terms of these.

2. Write $A = [\vec{a}_1 \ldots \vec{a}_6]$ as $A = [\vec{q}_1 \ldots \vec{q}_r]R$, where R *codes* each vector $\vec{a}_i$ in terms of the previously produced $\vec{q}_j$'s. Get $A = QR$. Check.

Matrix Inverses!

3. Check that C is a left-inverse of A (so $CA = I$) for:

$$A = \begin{bmatrix} -3 & -4 \\ 4 & 6 \\ 1 & 1 \end{bmatrix}$$

$$C = \frac{1}{2}\begin{bmatrix} 0 & -1 & 6 \\ 0 & 1 & -4 \end{bmatrix}$$

4. Let $\vec{b} = \begin{bmatrix} 1 \\ -2 \\ 0 \end{bmatrix}$. Write the equations that correspond to the system of linear equations

$$A\vec{x} = \vec{b}.$$

5. Use the fact that C is a left inverse of A to find a solution of this system. Check your solution.